MecHand: Freehand Chemical Mechanism Drawing with Sketch Recognition and Direct Export to LaTeX

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Abstract

Background: Reaction mechanisms are central to chemical education and research, yet existing tools for drawing them impose a steep learning curve or require expensive licences. Freehand sketching on paper remains the preferred medium for most chemists, but paper sketches cannot be directly integrated into digital manuscripts.

Results: We present **MecHand**, an open-source web application built on the Ketcher chemical drawing platform that accepts freehand input from stylus, trackpad or mouse and converts sketched mechanisms into clean, editable chemical diagrams. Stroke classification heuristics separate atom nodes, bonds and curved mechanism arrows before the sketch is rasterised and processed by MolScribe, a transformer-based molecular image recognition model. A "consolidating" interface presents the recognised structure to the user for confirmation or correction before it is committed to the canvas. MecHand exports diagrams directly to `chemfig` and TikZ (for LaTeX), as well as SVG and PNG. A context-sensitive help system provides tool descriptions calibrated for both expert chemists and undergraduate students.

Conclusions: MecHand fills a gap between the informality of hand sketching and the rigidity of graphical editors, lowering the barrier to producing publication-quality mechanism figures. The tool is freely available at https://

github.com/JordiVillaFreixa/Article_MecHand and runs entirely in the browser without installation, with an optional self-hosted backend for sketch recognition.

Keywords: reaction mechanism, chemical drawing, sketch recognition, MolScribe, chemfig, TikZ, open source

1 Background

Reaction mechanisms are routinely drawn by hand on paper or whiteboards during teaching and research discussions. Translating these sketches into digital manuscript-quality figures currently requires either expensive commercial software (e.g. ChemDraw, Marvin) or manual reconstruction in programmatic tools such as `chemfig` [1] or TikZ [2].

2 Implementation

2.1 Architecture overview

MecHand is structured as a three-layer system (Figure **??**):

1. **Frontend** — a fork of Ketcher [3] augmented with a freehand drawing overlay, the consolidating interface, a context-sensitive help system, and chemfig/TikZ export modules.
2. **Stroke classifier** — a lightweight geometric module that partitions pointer-event strokes into atom sites, bonds and mechanism arrows before recognition.
3. **Recognition backend** — a FastAPI service wrapping MolScribe [4] for molecular structure recognition and a contour-based arrow detector (Section 2.4).

2.2 Freehand input and stroke classification

2.3 The consolidating paradigm

2.4 Mechanism arrow detection

2.5 Export to chemfig and TikZ

2.6 Context-sensitive help system

3 Results and discussion

3.1 Benchmark on hand-drawn structures

We evaluated the recognition accuracy of MecHand on the DECIMER hand-drawn molecule dataset [5], which contains X structures scanned from student notebooks and published manuscripts.

3.2 User evaluation

4 Conclusions

5 Availability and requirements

- **Project name:** MecHand
- **Project home page:** https://github.com/JordiVillaFreixa/Article_MecHand
- **Operating system(s):** Platform independent (web application)
- **Programming language:** TypeScript (frontend), Python (backend)
- **Other requirements:** Modern browser (Chrome ≥110, Firefox ≥115, Safari ≥16); optional Python 3.10+ for the recognition backend.
- **License:** MIT (MecHand additions); Apache 2.0 (Ketcher upstream)
- **Any restrictions to use by non-academics:** None

Abbreviations

SVG: Scalable Vector Graphics; SMILES: Simplified Molecular-Input Line-Entry System; UX: user experience; API: application programming interface.

Declarations

Competing interests

The authors declare that they have no competing interests.

Funding

Authors' contributions

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